

Olympiad Test Paper — Maths (Class 7) — Paper 4

Chapter 15: Finding the Unknown

Paper 4 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	15 — Finding the Unknown
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: dropped signs while distributing nested brackets, averaging two rates instead of total-distance-over-total-time, off-by-one in consecutive/digit set-ups, reporting the complement of the unknown, and stopping at x instead of the expression actually asked. Do not write on this paper — mark answers on your answer sheet.

1. Solve: $3[2x - (x - 5)] - 4 = 2(x + 6)$.

- (A) 1 (B) -1
(C) 11 (D) 19

2. Solve: $4x - 2[5 - (x - 3)] = x - 6$.

- (A) -2 (B) 4
(C) 2 (D) 10

3. Solve: $7x - \{3 - [2 - (x - 4)]\} = 5x + 1$.

- (A) -2 (B) 2
(C) -6 (D) 6

4. How many values of x satisfy $4(2x - 1) - 3[x - (2 - x)] = 2x + 5$?

- (A) exactly one (B) none
(C) infinitely many (every number) (D) exactly two

5. Solve: $6 - \{2x - [4 - (x - 1)]\} = 3 - x$.

- (A) -4 (B) 2
(C) 8 (D) 4

6. Solve: $5 - \{3 - [2x - (4 - x)]\} = 2x + 1$.

- (A) -3 (B) 1
(C) 9 (D) 3

7. Solve: $3[x - 2(x - 3)] = 4 - \{x - [2 - (x - 1)]\}$.

- (A) -11 (B) 5
(C) 11 (D) 7

8. How many values of x satisfy $3(2x - 1) - 2[4 - (3 - x)] = 4x - 5$?

- (A) exactly one (B) none
(C) infinitely many (every number) (D) exactly two

9. Solve: $(2x + 3)/5 - (x - 1)/2 = (x + 7)/10$.

- (A) -2 (B) 3
(C) 2 (D) 12

10. Solve: $(x - 1)/(2x + 1) = 2/5$.

- (A) -7 (B) 7
(C) 3 (D) 9

11. Solve: $(x - 1)/2 + (2x + 1)/3 = (5x - 1)/6 + 2$.

- (A) -6 (B) 2
(C) 6 (D) 4

12. Solve: $3/(2x - 1) = 4/(3x + 1)$.

- (A) -7 (B) 7
(C) 5 (D) 1

13. Two-fifths of a number is 6 less than one-half of the same number. The number is:

- (A) 60 (B) 20
(C) 30 (D) 12

14. A fraction has denominator 3 more than its numerator. When 1 is subtracted from both numerator and denominator, it becomes $3/4$. The original fraction is:

- (A) $7/10$ (B) $9/12$
(C) $10/13$ (D) $11/14$

15. Solve: $x/2 + x/3 - x/4 = 7$.

- (A) 12 (B) 6
(C) 21 (D) 28

16. A father is 5 times as old as his son now. In 6 years he will be only 3 times as old. The son's present age (in years) is:

- (A) 12 (B) 18
(C) 9 (D) 6

17. The sum of the present ages of a mother and her daughter is 50 years. Five years ago the mother was 7 times as old as her daughter. The daughter's present age (in years) is:

- | | |
|-------|--------|
| (A) 5 | (B) 15 |
| (C) 8 | (D) 10 |

18. The sum of the present ages of A and B is 60 years. Ten years ago A was 3 times as old as B. A's present age (in years) is:

- | | |
|--------|--------|
| (A) 20 | (B) 45 |
| (C) 40 | (D) 30 |

19. The present ages of two cousins are in the ratio 5 : 2. After 6 years the ratio will be 2 : 1. The younger cousin's present age (in years) is:

- | | |
|--------|--------|
| (A) 6 | (B) 12 |
| (C) 30 | (D) 18 |

20. A man's age is 3 years more than twice his son's age. In 4 years the sum of their ages will be 56. The man's present age (in years) is:

- | | |
|--------|--------|
| (A) 33 | (B) 15 |
| (C) 30 | (D) 36 |

21. A man is 24 years older than his daughter. Six years from now he will be 3 times as old as she will be. The daughter's present age (in years) is:

- | | |
|--------|-------|
| (A) 12 | (B) 6 |
| (C) 18 | (D) 8 |

22. A two-digit number equals 7 times its units digit, and its tens digit is 2 less than its units digit. The number is:

- | | |
|--------|--------|
| (A) 57 | (B) 46 |
| (C) 53 | (D) 35 |

23. The sum of four consecutive integers is 2 less than 3 times the largest of them. The smallest integer is:

- | | |
|--------|-------|
| (A) -1 | (B) 3 |
| (C) 2 | (D) 1 |

24. Reversing the digits of a two-digit number increases it by 36, and the sum of its digits is 10. The original number is:

- | | |
|--------|--------|
| (A) 73 | (B) 37 |
| (C) 28 | (D) 46 |

25. The sum of three consecutive multiples of 5 equals 5 times the smallest, plus 5. The largest of them is:

- | | |
|--------|--------|
| (A) 5 | (B) 10 |
| (C) 20 | (D) 15 |

26. A fraction equals $\frac{3}{5}$. When 2 is added to its numerator and 2 is subtracted from its denominator, the value becomes 1. The original fraction is:

- (A) $\frac{6}{10}$ (B) $\frac{3}{5}$
(C) $\frac{9}{15}$ (D) $\frac{4}{8}$

27. Of three consecutive even integers, the largest is 6 less than twice the smallest. The middle integer is:

- (A) 10 (B) 14
(C) 8 (D) 12

28. The angles of a triangle are $(2x)^\circ$, $(3x - 10)^\circ$ and $(x + 10)^\circ$. The largest angle measures:

- (A) 80° (B) 60°
(C) 40° (D) 90°

29. A rectangle has length $(3x - 2)$ cm and breadth $(x + 1)$ cm. The length is 7 cm more than the breadth. The perimeter (in cm) is:

- (A) 19 (B) 38
(C) 42 (D) 26

30. Each interior angle of a regular polygon is 150° . The number of sides is:

- (A) 10 (B) 12
(C) 15 (D) 8

31. An isosceles triangle has perimeter 50 cm; its base is 5 cm longer than each equal side. Each equal side (in cm) is:

- (A) 20 (B) 15
(C) 12 (D) 10

32. The angles of a quadrilateral are in the ratio 3 : 4 : 5 : 6. The largest angle measures:

- (A) 100° (B) 108°
(C) 144° (D) 120°

33. In a triangle each of two equal angles is 15° more than the third angle. The third (smallest) angle measures:

- (A) 50° (B) 65°
(C) 40° (D) 45°

34. A bag has ₹1, ₹2 and ₹5 coins in the ratio 4 : 3 : 2, worth ₹120 in all. The number of ₹5 coins is:

- (A) 18 (B) 24
(C) 6 (D) 12

35. A theatre sold 50 tickets for ₹2350. Adult tickets cost ₹60 and child tickets ₹35. The number of adult tickets sold was:

- (A) 26 (B) 20
(C) 24 (D) 30

36. An 80-litre mixture has milk and water in the ratio 5 : 3. How much water must be added to make the ratio 1 : 1?

- (A) 10 L (B) 16 L
(C) 20 L (D) 30 L

37. ₹560 is shared among A, B and C so that A gets ₹40 more than B, and C gets twice B's share. C's share (in ₹) is:

- (A) 130 (B) 170
(C) 260 (D) 320

38. A shopkeeper sold 36 articles, some at ₹25 and the rest at ₹40, taking ₹1140 in all. The number sold at ₹40 was:

- (A) 20 (B) 18
(C) 24 (D) 16

39. A box has only ₹2 and ₹5 coins. The number of ₹2 coins is 4 more than twice the number of ₹5 coins, and the box holds ₹98. The number of ₹2 coins is:

- (A) 10 (B) 14
(C) 20 (D) 24

40. A car covers the first 120 km of a journey at 40 km/h and the next 120 km at 60 km/h. Its average speed for the whole journey (km/h) is:

- (A) 48 (B) 50
(C) 100 (D) 52

41. Pipe A fills a tank in 4 hours and pipe B in 6 hours. Working together, the time (in hours) to fill the tank is:

- (A) 2.4 (B) 5
(C) 10 (D) 3

42. A boat covers 30 km downstream in 2 hours and the same 30 km upstream in 5 hours. The speed of the stream (in km/h) is:

- (A) 9 (B) 4.5
(C) 6 (D) 10.5

43. A train passes a pole in 12 seconds and a 120-metre platform in 24 seconds at the same speed. The length of the train (in metres) is:

- (A) 120 (B) 240
(C) 60 (D) 180

44. Two cars start 240 km apart and drive toward each other at 45 km/h and 35 km/h. The time (in hours) until they meet is:

(A) 6.857...

(B) 3

(C) 24

(D) 5.333...

45. A man rows 18 km downstream and the same 18 km back upstream. Downstream his speed is 9 km/h and upstream 6 km/h. His total time for the round trip (in hours) is:

(A) 5

(B) 4.8

(C) 2.4

(D) 3

46. If $5x - 4 = 3x + 8$, then the value of $x^2 + x$ is:

(A) 6

(B) 42

(C) 36

(D) 30

47. If $4(x - 1) = 2(x + 3)$, then the value of $3x^2 - 5$ is:

(A) 5

(B) 25

(C) 70

(D) 20

48. For which value of k does the equation $2(x + k) = 3x - 7$ have the solution $x = 5$?

(A) -1

(B) 1

(C) -6

(D) 6

49. If $7x - 3 = 4x + 9$, then the value of $2x + 5$ is:

(A) 4

(B) 8

(C) 13

(D) 17

50. If $6x - 5 = 4x + 7$, then the value of $(x^2 - 4)/2$ is:

(A) 6

(B) 32

(C) 18

(D) 16

51. For which value of m does the equation $m(x - 1) = 2x - 2$ hold for EVERY value of x ?

(A) 1

(B) -2

(C) 2

(D) 0

52. Translate: 'Twice the sum of a number n and 3 is 4 less than five times the number.'

The equation is:

(A) $2n + 3 = 5n - 4$

(B) $2(n + 3) = 5n + 4$

(C) $2(n + 3) = 5n - 4$

(D) $2(n + 3) = 4 - 5n$

53. Three pens and two pencils cost ₹72, while two pens and three pencils cost ₹68. The cost of one pen (in ₹) is:

(A) 16

(B) 12

(C) 14

(D) 20

54. Five kilograms of apples and two kilograms of grapes cost ₹430, while three kilograms of apples and four kilograms of grapes cost ₹440. The price per kilogram of apples (in ₹) is:

(A) 65

(B) 70

(C) 55

(D) 60

55. Three-fourths of a number exceeds two-thirds of the same number by 5. The number is:

(A) 12

(B) 60

(C) 30

(D) 20

56. In a 40-question test, +5 is awarded for each correct answer and -2 for each wrong answer, with all questions attempted. A student scores 102. The number of correct answers is:

(A) 22

(B) 26

(C) 28

(D) 20

57. If $2x + 3y = 12$ and $x - y = 1$, then the value of y is:

(A) 3

(B) 2

(C) 4

(D) 1

58. Two numbers are in the ratio 5 : 7. If 8 is added to each, the new ratio is 7 : 9. The smaller number is:

(A) 28

(B) 12

(C) 20

(D) 16

59. If $3a - 2b = 5$ and $a + 2b = 11$, then the value of a is:

(A) 3

(B) 16

(C) 7

(D) 4

60. A man spends one-third of his money on rent and one-fourth of it on food, and is left with ₹250. His total money (in ₹) was:

(A) 500

(B) 600

(C) 750

(D) 450

Solutions — Maths (Class 7), Chapter 15: Finding the Unknown — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	C	21	B	31	B	41	A	51	C
2	C	12	A	22	D	32	D	42	B	52	C
3	A	13	A	23	D	33	A	43	A	53	A
4	B	14	C	24	B	34	D	44	B	54	D
5	D	15	A	25	D	35	C	45	A	55	B
6	D	16	D	26	A	36	C	46	B	56	B
7	C	17	D	27	D	37	C	47	C	57	B
8	C	18	C	28	A	38	D	48	A	58	C
9	C	19	B	29	B	39	D	49	C	59	D
10	B	20	A	30	B	40	A	50	D	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Inner bracket: $2x - (x - 5) = 2x - x + 5 = x + 5$. Then $3(x + 5) - 4 = 3x + 15 - 4 = 3x + 11$. The equation $3x + 11 = 2x + 12$ gives $3x - 2x = 12 - 11$, so $x = 1$. Check: $3[2 - (-4)] - 4 = 3(6) - 4 = 14$ and $2(7) = 14$. Forgetting to flip the sign inside the bracket (using $x - 5$, so $3(x - 5) - 4 = 2x + 12$) gives -1; leaving the bracket un-tripled ($x + 11 = 2x + 12$) gives 11; mis-combining $15 + 4$ instead of $15 - 4$ in the constant gives 19.

2. (C) Inner: $5 - (x - 3) = 5 - x + 3 = 8 - x$. Then $-2(8 - x) = -16 + 2x$, so the left side is $4x - 16 + 2x = 6x - 16$. The equation $6x - 16 = x - 6$ gives $5x = 10$, so $x = 2$. Check: $4(2) - 2[5 - (-1)] = 8 - 2(6) = 8 - 12 = -4$ and $2 - 6 = -4$. Failing to flip the inner sign (using $5 - x - 3 = 2 - x$) corrupts the constant toward 4; distributing -2 as $-2x$ gives a sign trap -2; reading $5x = 10$ as $5x = 50$ gives 10.

3. (A) Build outward. Innermost: $x - 4$. Next: $2 - (x - 4) = 2 - x + 4 = 6 - x$. Next: $3 - [6 - x] = 3 - 6 + x = x - 3$. So the left side is $7x - (x - 3) = 7x - x + 3 = 6x + 3$. The equation $6x + 3 = 5x + 1$ gives $x = -2$. Check: $7(-2) - \{3 - [2 - (-6)]\} = -14 - \{3 - 8\} = -14 + 5 = -9$

and $5(-2) + 1 = -9$. Mishandling one nested sign gives +2 or 6; dropping the outer brace minus gives -6.

4. (B) Simplify the bracket: $x - (2 - x) = x - 2 + x = 2x - 2$, so $3(2x - 2) = 6x - 6$. The left side is $8x - 4 - (6x - 6) = 8x - 4 - 6x + 6 = 2x + 2$. The equation becomes $2x + 2 = 2x + 5 \rightarrow 2 = 5$, which is impossible, so there is NO solution. Finding one root and stopping suggests 'exactly one'; reading the two constants as equal suggests 'every number'; a linear equation can never have 'exactly two' roots.

5. (D) Innermost: $x - 1$. Next: $4 - (x - 1) = 4 - x + 1 = 5 - x$. Next: $2x - [5 - x] = 2x - 5 + x = 3x - 5$. Then $6 - (3x - 5) = 6 - 3x + 5 = 11 - 3x$. The equation $11 - 3x = 3 - x$ gives $11 - 3 = 3x - x \rightarrow 8 = 2x \rightarrow x = 4$. Check: $6 - \{8 - [4 - 3]\} = 6 - \{8 - 1\} = 6 - 7 = -1$ and $3 - 4 = -1$. A sign slip on the inner bracket gives -4 or 8; mis-transposing gives 2.

6. (D) Innermost: $4 - x$. Next: $2x - (4 - x) = 2x - 4 + x = 3x - 4$. Next: $3 - [3x - 4] = 3 - 3x + 4 = 7 - 3x$. Then $5 - (7 - 3x) = 5 - 7 + 3x = 3x - 2$. The equation $3x - 2 = 2x + 1$ gives $x = 3$. Check: $5 - \{3 - [6 - 1]\} = 5 - \{3 - 5\} = 5 + 2 = 7$ and $2(3) + 1 = 7$. Not flipping a nested sign gives -3 or 9; a constant slip gives 1.

7. (C) Left side: $x - 2(x - 3) = x - 2x + 6 = -x + 6$, so $3(-x + 6) = -3x + 18$. Right side innermost: $x - 1$; then $2 - (x - 1) = 3 - x$; then $x - [3 - x] = x - 3 + x = 2x - 3$; then $4 - [2x - 3] = 4 - 2x + 3 = 7 - 2x$. The equation $-3x + 18 = 7 - 2x$ gives $18 - 7 = 3x - 2x \rightarrow x = 11$. Check: $3[11 - 2(8)] = 3[11 - 16] = -15$ and $4 - \{11 - [2 - 10]\} = 4 - \{11 + 8\} = 4 - 19 = -15$. Sign mishandling of the nested terms gives -11, 5, or 7.

8. (C) Simplify: $4 - (3 - x) = 4 - 3 + x = 1 + x$, so $2(1 + x) = 2 + 2x$. The left side is $6x - 3 - (2 + 2x) = 6x - 3 - 2 - 2x = 4x - 5$, which is identical to the right side $4x - 5$. The statement $4x - 5 = 4x - 5$ holds for every value of x , so there are infinitely many solutions. Solving and reading off a single root suggests 'exactly one'; treating the equal constants as unequal suggests 'none'; a linear identity is never 'exactly two'.

9. (C) Multiply every term by the LCD 10: $2(2x + 3) - 5(x - 1) = (x + 7)$. Expand: $4x + 6 - 5x + 5 = x + 7 \rightarrow -x + 11 = x + 7 \rightarrow 11 - 7 = x + x \rightarrow 4 = 2x \rightarrow x = 2$. Verify: $(7)/5 - (1)/2 = 1.4 - 0.5 = 0.9$ and $(9)/10 = 0.9$. Distributing $-5(x - 1)$ as $-5x - 5$ corrupts the constant toward -2; a transposition slip gives 3; clearing only one denominator gives 12.

10. (B) Cross-multiply: $5(x - 1) = 2(2x + 1) \rightarrow 5x - 5 = 4x + 2 \rightarrow 5x - 4x = 2 + 5 \rightarrow x = 7$. Check: $(6)/(15) = 2/5$. Cross-multiplying the wrong way, $2(x - 1) = 5(2x + 1)$, gives a negative trap -7; transposition slips give 3 or 9.

11. (C) Multiply every term by the LCD 6: $3(x - 1) + 2(2x + 1) = (5x - 1) + 12$. Expand: $3x - 3 + 4x + 2 = 5x + 11 \rightarrow 7x - 1 = 5x + 11 \rightarrow 2x = 12 \rightarrow x = 6$. Verify: $(5)/2 + (13)/3 = 2.5 + 4.333... = 6.833...$ and $(29)/6 + 2 = 4.833... + 2 = 6.833...$. A sign error on $(x - 1)$ gives -6; mis-clearing the LCD gives 2 or 4.

12. (A) Cross-multiply: $3(3x + 1) = 4(2x - 1) \rightarrow 9x + 3 = 8x - 4 \rightarrow 9x - 8x = -4 - 3 \rightarrow x = -7$. Check: $3/(-15) = -1/5$ and $4/(-20) = -1/5$. A dropped minus gives 7; transposition slips give 5 or 1.

13. (A) Let the number be n . The condition is $(1/2)n = (2/5)n + 6$, i.e. half exceeds two-fifths by 6: $(1/2)n - (2/5)n = 6$. The LCD is 10: $(5n - 4n)/10 = 6 \rightarrow n/10 = 6 \rightarrow n = 60$. Check: half of 60 is 30, two-fifths is 24, and $30 - 24 = 6$. Equating $(2/5)n - (1/2)n = 6$ reverses the gap and gives a negative; using a difference of one-tenth as one-fifth gives 30; reading 'two-fifths' as the whole gap gives 20 or 12.

14. (C) Let the numerator be n , so the denominator is $n + 3$ and the fraction is $n/(n + 3)$. Subtracting 1 from each: $(n - 1)/(n + 2) = 3/4$. Cross-multiply: $4(n - 1) = 3(n + 2) \rightarrow 4n - 4 = 3n + 6 \rightarrow n = 10$. The denominator is 13, so the fraction is $10/13$. Check: $(9)/(12) = 3/4$. Mis-placing the gap of 3 gives $7/10$; an off-by-one in n gives $9/12$ or $11/14$.

15. (A) Multiply every term by the LCD 12: $6x + 4x - 3x = 84 \rightarrow 7x = 84 \rightarrow x = 12$. Check: $6 + 4 - 3 = 7$. Adding all three coefficients as $6 + 4 + 3 = 13$ mis-signs the $-x/4$ and shifts the root; halving 12 gives 6; multiplying 7 by 3 or 4 instead of dividing gives 21 or 28.

16. (D) Let the son be s , so the father is $5s$. In 6 years: son $s + 6$, father $5s + 6$, with $5s + 6 = 3(s + 6)$. Expand: $5s + 6 = 3s + 18 \rightarrow 5s - 3s = 18 - 6 \rightarrow 2s = 12 \rightarrow s = 6$. Check: son 6, father 30; in 6 years 12 and 36, and $36 = 3 \times 12$. Comparing present ages $5s = 3s$ gives a nonsense; forgetting to add 6 to the son gives $5s + 6 = 3s \rightarrow$ negative, mis-corrected to 12 or 18; halving the wrong gap gives 9.

17. (D) Let the daughter be d , so the mother is $50 - d$. Five years ago: daughter $d - 5$, mother $45 - d$, with $45 - d = 7(d - 5)$. Expand: $45 - d = 7d - 35 \rightarrow 45 + 35 = 7d + d \rightarrow 80 = 8d \rightarrow d = 10$. Check: daughter 10, mother 40; five years ago 5 and 35, and $35 = 7 \times 5$. Forgetting to subtract 5 from the daughter gives a smaller 5; mis-summing gives 15; a sign slip gives 8.

18. (C) Let B be b , so A is $60 - b$. Ten years ago: $A = 50 - b$, $B = b - 10$, with $50 - b = 3(b - 10)$. Expand: $50 - b = 3b - 30 \rightarrow 50 + 30 = 3b + b \rightarrow 80 = 4b \rightarrow b = 20$, so $A = 40$. Check: ten years ago $A = 40$, $B = 10$, and $40 = 3 \times 10$. Reporting B 's age gives 20; forgetting to subtract 10 from B gives 45; halving the sum gives 30.

19. (B) Let the ages be $5k$ and $2k$. After 6 years: $(5k + 6) : (2k + 6) = 2 : 1$, so $5k + 6 = 2(2k + 6) \rightarrow 5k + 6 = 4k + 12 \rightarrow k = 6$. The younger is $2k = 12$ (and the elder 30). Check: in 6 years 36 and 18, and $36 : 18 = 2 : 1$. Reporting k gives 6; reporting the elder gives 30; mis-scaling gives 18.

20. (A) Let the son be s , so the man is $2s + 3$. In 4 years the sum is $(2s + 3 + 4) + (s + 4) = 3s + 11 = 56 \rightarrow 3s = 45 \rightarrow s = 15$, so the man is $2(15) + 3 = 33$. Check: man 33, son 15; in 4 years 37 and 19, sum 56. Reporting the son's age gives 15; forgetting the $+3$ gives 30; mis-handling the $+4$'s gives 36.

21. (B) Let the daughter be d , so the man is $d + 24$. In 6 years: daughter $d + 6$, man $d + 30$, with $d + 30 = 3(d + 6)$. Expand: $d + 30 = 3d + 18 \rightarrow 30 - 18 = 3d - d \rightarrow 12 = 2d \rightarrow d = 6$. Check: daughter 6, man 30; in 6 years 12 and 36, and $36 = 3 \times 12$. Comparing present ages $d + 24 = 3d$ gives 12; forgetting to add 6 to the daughter gives 18; a halving slip gives 8.

22. (D) Let the tens digit be t and units u , number $10t + u$. The condition $10t + u = 7u$ gives $10t = 6u \rightarrow 5t = 3u$. With $t = u - 2$: $5(u - 2) = 3u \rightarrow 5u - 10 = 3u \rightarrow 2u = 10 \rightarrow u = 5$, $t = 3$, number 35. Check: $7 \times 5 = 35$ and $3 = 5 - 2$. Reversing the relation gives 53; ignoring $5t = 3u$ gives 57 or 46.

23. (D) Let the integers be $n, n + 1, n + 2, n + 3$. Their sum is $4n + 6$, and 3 times the largest minus 2 is $3(n + 3) - 2 = 3n + 7$. Set $4n + 6 = 3n + 7 \rightarrow n = 1$, giving 1, 2, 3, 4. Check: sum = 10 and $3 \times 4 - 2 = 10$. Reporting the second term gives 2; a sign slip on the constants gives -1 ; using the largest gives 3.

24. (B) Let tens t , units u , number $10t + u$. Reversed $10u + t = (10t + u) + 36 \rightarrow 9u - 9t = 36 \rightarrow u - t = 4$. With $u + t = 10$: adding gives $2u = 14 \rightarrow u = 7$, $t = 3$, number 37. Check: reversed $73 = 37 + 36$ and $3 + 7 = 10$. The reversed number 73 is a trap; mis-solving the pair gives 28 or 46.

25. (D) Let the multiples be $5n, 5n + 5, 5n + 10$. Their sum is $15n + 15$, and 5 times the smallest plus 5 is $5(5n) + 5 = 25n + 5$. Set $15n + 15 = 25n + 5 \rightarrow 15 - 5 = 25n - 15n \rightarrow 10 = 10n \rightarrow n = 1$, giving 5, 10, 15; the largest is 15. Check: sum $30 = 25(1) + 5$. Reporting the smallest gives 5, the middle 10; adding 5 again gives 20.

26. (A) Since the fraction equals $3/5$, write it as $3k/5k$. After the changes: $(3k + 2)/(5k - 2) = 1$, so $3k + 2 = 5k - 2 \rightarrow 4 = 2k \rightarrow k = 2$, giving $6/10$. Check: $(6 + 2)/(10 - 2) = 8/8 = 1$. The lowest-terms form $3/5$ fails the test ($5/3 \neq 1$); $k = 3$ gives $9/15$ which fails ($(11)/(13) \neq 1$); $4/8$ is not equal to $3/5$.

27. (D) Let them be $n, n + 2, n + 4$. The condition is $n + 4 = 2n - 6 \rightarrow 4 + 6 = 2n - n \rightarrow n = 10$, giving 10, 12, 14; the middle is 12. Check: $14 = 2(10) - 6$. Reporting the smallest gives 10, the largest gives 14; mis-solving gives 8.

28. (A) Angles sum to 180° : $2x + (3x - 10) + (x + 10) = 180 \rightarrow 6x = 180 \rightarrow x = 30$. The angles are $60^\circ, 80^\circ, 40^\circ$, so the largest is 80° . Check: $60 + 80 + 40 = 180$. Reporting $(2x)$ gives 60° and $(x + 10)$ gives 40° ; mis-summing the constants to make $6x = 180 \pm 20$ gives 90° .

29. (B) The length exceeds the breadth by 7: $(3x - 2) - (x + 1) = 7 \rightarrow 2x - 3 = 7 \rightarrow 2x = 10 \rightarrow x = 5$. Then length = 13 cm, breadth = 6 cm, perimeter = $2(13 + 6) = 38$ cm. Check: $13 - 6 = 7$. Using one length-plus-breadth without doubling gives 19; mis-solving x gives 42 or 26.

30. (B) Each exterior angle $= 180^\circ - 150^\circ = 30^\circ$, and exterior angles sum to 360° , so the number of sides $= 360 \div 30 = 12$. Misreading $180 - 150$ as 36 gives 10; dividing 360 by 24 gives 15; by 45 gives 8.

31. (B) Let each equal side be s , so the base is $s + 5$. Perimeter: $2s + (s + 5) = 50 \rightarrow 3s + 5 = 50 \rightarrow 3s = 45 \rightarrow s = 15$. The base is 20. Check: $15 + 15 + 20 = 50$. Reporting the base gives 20; treating the base as 5 less than the side gives 12; splitting 45 by halves instead of thirds gives 10.

32. (D) The four parts sum to $3 + 4 + 5 + 6 = 18$, and a quadrilateral's angles total 360° , so one part $= 360 \div 18 = 20^\circ$. The largest angle is $6 \times 20 = 120^\circ$. Check: $(3 + 4 + 5 + 6) \times 20 = 360$. Using $350 \div 18$ mis-rounds toward 100° ; dividing by a triangle's $540 \div 18 \times 6$ idea gives 108° or 144° .

33. (A) Let the third angle be t , so each equal angle is $t + 15$. Then $t + 2(t + 15) = 180 \rightarrow 3t + 30 = 180 \rightarrow 3t = 150 \rightarrow t = 50$. Check: $50 + 65 + 65 = 180$. Reporting an equal angle gives 65° ; subtracting 15 from the third gives 40° or 45° .

34. (D) Let the counts be $4k, 3k, 2k$. The value is $1(4k) + 2(3k) + 5(2k) = 4k + 6k + 10k = 20k = 120 \rightarrow k = 6$. The ₹5 coins number $2k = 12$. Check: $24 + 36 + 60 = ₹120$ with 24, 18, 12 coins. Reading the ratio number for ₹1 coins gives $4k = 24$; for ₹2 coins $3k = 18$; halving k gives 6.

35. (C) Let a be adult tickets, so child tickets number $50 - a$. Value: $60a + 35(50 - a) = 2350 \rightarrow 60a + 1750 - 35a = 2350 \rightarrow 25a = 600 \rightarrow a = 24$. Check: 24 adults = ₹1440, 26 children = ₹910, total ₹2350. The complement $50 - 24 = 26$ is the child count, a swap trap; splitting evenly gives 25 toward 20 or 30.

36. (C) In 80 L the milk is $(5/8) \times 80 = 50$ L and water $(3/8) \times 80 = 30$ L. Adding w litres of water keeps milk at 50 L; for 1 : 1 we need water = milk, so $30 + w = 50 \rightarrow w = 20$. Check: 50 L milk and 50 L water is 1 : 1. Halving the gap gives 10 L; using the original 16-L difference idea gives 16 L; matching the milk total itself gives 30 L.

37. (C) Let $B = b$. Then $A = b + 40$ and $C = 2b$, summing to 560: $(b + 40) + b + 2b = 560 \rightarrow 4b + 40 = 560 \rightarrow 4b = 520 \rightarrow b = 130$. So $C = 2b = 260$. Check: $A = 170, B = 130, C = 260$, total 560. Reporting B gives 130; reporting A gives 170; doubling the total share idea gives 320.

38. (D) Let h be the ₹40 articles, so ₹25 articles number $36 - h$. Value: $40h + 25(36 - h) = 1140 \rightarrow 40h + 900 - 25h = 1140 \rightarrow 15h = 240 \rightarrow h = 16$. Check: $16 \times 40 = 640$ and $20 \times 25 = 500$, total 1140. The complement $36 - 16 = 20$ is the ₹25 count, a swap trap; splitting evenly gives 18; a sign slip gives 24.

39. (D) Let the ₹5 coins number f , so the ₹2 coins number $2f + 4$. Value: $5f + 2(2f + 4) = 98 \rightarrow 5f + 4f + 8 = 98 \rightarrow 9f = 90 \rightarrow f = 10$, so ₹2 coins $= 2(10) + 4 = 24$. Check: $10 \times 5 =$

₹50 and $24 \times 2 = ₹48$, total ₹98. Reporting the ₹5 count gives 10; forgetting the +4 gives 20; using 'twice f' without the +4 then mis-adding gives 14.

40. (A) Average speed is total distance over total time, never the mean of the speeds. First leg time = $120/40 = 3$ h; second leg = $120/60 = 2$ h; total 240 km in 5 h, so average = $240/5 = 48$ km/h. The naive mean $(40 + 60)/2 = 50$ is the trap; adding the speeds gives 100; a time miscount gives 52.

41. (A) Rates add, not times. In one hour A fills $1/4$ and B fills $1/6$, together $1/4 + 1/6 = 3/12 + 2/12 = 5/12$ of the tank, so the whole tank takes $12/5 = 2.4$ hours. Averaging the times $(4 + 6)/2 = 5$ is the trap; adding the times gives 10; a rate slip gives 3.

42. (B) Downstream speed = $30/2 = 15$ km/h = boat + stream; upstream speed = $30/5 = 6$ km/h = boat – stream. Subtracting, $2 \times \text{stream} = 15 - 6 = 9$, so stream = 4.5 km/h (and boat = 10.5). The boat speed 10.5 is a trap; the downstream speed 15 $\rightarrow$ boat 9 is another; using only one leg gives 6.

43. (A) Past the pole the train covers its own length L in 12 s, so speed = $L/12$. Past the platform it covers $L + 120$ in 24 s, so speed = $(L + 120)/24$. Equate: $L/12 = (L + 120)/24 \rightarrow 24L = 12(L + 120) \rightarrow 24L = 12L + 1440 \rightarrow 12L = 1440 \rightarrow L = 120$. Check: speed 10 m/s; pole $120/10 = 12$ s, platform $240/10 = 24$ s. Doubling gives 240; halving gives 60; misadding gives 180.

44. (B) Moving toward each other, the closing speed is the sum $45 + 35 = 80$ km/h, so the time to close 240 km is $240/80 = 3$ hours. Using only one speed gives $240/35 \approx 6.857$; using the difference 10 km/h gives 24; using $240/45$ gives about 5.333.

45. (A) Time is distance over speed for each leg. Downstream: $18/9 = 2$ h; upstream: $18/6 = 3$ h; total = $2 + 3 = 5$ h. Using the average speed $(9 + 6)/2 = 7.5$ km/h on 36 km gives $36/7.5 = 4.8$ h, the classic averaging trap; using one leg's speed on 36 km gives values like 4 or, via $36/15$, 2.4; taking only the upstream leg gives 3.

46. (B) Solve: $5x - 4 = 3x + 8 \rightarrow 5x - 3x = 8 + 4 \rightarrow 2x = 12 \rightarrow x = 6$. Then $x^2 + x = 36 + 6 = 42$. Stopping at $x = 6$ reports 6; computing only x^2 gives 36; using $5x$ without context gives 30.

47. (C) Expand: $4x - 4 = 2x + 6 \rightarrow 4x - 2x = 6 + 4 \rightarrow 2x = 10 \rightarrow x = 5$. Then $3x^2 - 5 = 3(25) - 5 = 70$. Reporting x itself gives 5; computing $(3x)^2$ loosely or 5^2 gives 25; using $3x - 5 = 10$ gives 20.

48. (A) Substitute $x = 5$: $2(5 + k) = 3(5) - 7 \rightarrow 10 + 2k = 8 \rightarrow 2k = -2 \rightarrow k = -1$. Verify: $2(5 - 1) = 8$ and $15 - 7 = 8$, equal. A halving slip gives 1; forgetting to halve gives ± 6 .

49. (C) Solve: $7x - 3 = 4x + 9 \rightarrow 3x = 12 \rightarrow x = 4$. Then $2x + 5 = 8 + 5 = 13$. Stopping at $x = 4$ reports 4; computing only $2x$ gives 8; using $3x + 5$ gives 17.

50. (D) Solve: $6x - 5 = 4x + 7 \rightarrow 2x = 12 \rightarrow x = 6$. Then $(x^2 - 4)/2 = (36 - 4)/2 = 32/2 = 16$. Reporting x itself reports 6; forgetting to divide by 2 gives 32; dropping the -4 and computing $x^2/2 = 36/2$ gives 18.

51. (C) The right side factors as $2x - 2 = 2(x - 1)$. The equation $m(x - 1) = 2(x - 1)$ is true for every x exactly when the coefficients match, i.e. $m = 2$ (then both sides are identical). Check: $2(x - 1) = 2x - 2$ for all x . Taking $m = 1$ leaves $x - 1 = 2x - 2$, which holds only at $x = 1$, not everywhere; $m = -2$ or $m = 0$ makes the two sides differ for almost all x .

52. (C) 'The sum of n and 3' is $(n + 3)$; 'twice the sum' multiplies the whole bracket: $2(n + 3)$. '4 less than five times the number' is $5n - 4$. So $2(n + 3) = 5n - 4$. Multiplying only n by 2 drops the bracket; '4 less' is -4 , not $+4$; '4 less than $5n$ ' is $5n - 4$, not $4 - 5n$.

53. (A) Let a pen cost p and a pencil q . Then $3p + 2q = 72$ and $2p + 3q = 68$. Multiply the first by 3 and the second by 2: $9p + 6q = 216$ and $4p + 6q = 136$. Subtract: $5p = 80 \rightarrow p = 16$ (and $q = 12$). Check: $3(16) + 2(12) = 72$ and $2(16) + 3(12) = 68$. The pencil price 12 is a swap trap; arithmetic slips give 14 or 20.

54. (D) Let apples cost a and grapes g per kg. Then $5a + 2g = 430$ and $3a + 4g = 440$. Multiply the first by 2: $10a + 4g = 860$. Subtract the second: $7a = 420 \rightarrow a = 60$ (and $g = 65$). Check: $5(60) + 2(65) = 430$ and $3(60) + 4(65) = 440$. The grape price 65 is a swap trap; mis-elimination gives 70 or 55.

55. (B) Let the number be n . The condition is $(3/4)n - (2/3)n = 5$. The LCD is 12: $(9n - 8n)/12 = 5 \rightarrow n/12 = 5 \rightarrow n = 60$. Check: three-fourths of 60 is 45, two-thirds is 40, and $45 - 40 = 5$. Treating the difference $(1/12)$ as $(1/5)$ gives 25 toward 20; misreading the gap gives 12 or 30.

56. (B) Let correct = c , so wrong = $40 - c$. Score: $5c - 2(40 - c) = 102 \rightarrow 5c - 80 + 2c = 102 \rightarrow 7c = 182 \rightarrow c = 26$. Check: 26 correct gives +130, 14 wrong gives -28 , net 102. Forgetting the penalty ($5c = 102$) drifts toward 20; a sign error on the -2 gives 22 or 28.

57. (B) From $x - y = 1$, $x = y + 1$. Substitute: $2(y + 1) + 3y = 12 \rightarrow 2y + 2 + 3y = 12 \rightarrow 5y = 10 \rightarrow y = 2$ (and $x = 3$). Check: $2(3) + 3(2) = 12$ and $3 - 2 = 1$. The value of x , 3, is a swap trap; a mis-substitution gives 4 or 1.

58. (C) Let the numbers be $5k$ and $7k$. Then $(5k + 8) : (7k + 8) = 7 : 9$, so $9(5k + 8) = 7(7k + 8) \rightarrow 45k + 72 = 49k + 56 \rightarrow 72 - 56 = 49k - 45k \rightarrow 16 = 4k \rightarrow k = 4$. The smaller number is $5k = 20$ (and the larger 28). Check: $28 : 36 = 7 : 9$. Reporting the larger gives 28; using k itself gives 4 toward 12 or 16.

59. (D) Add the two equations to eliminate b : $(3a - 2b) + (a + 2b) = 5 + 11 \rightarrow 4a = 16 \rightarrow a = 4$ (and $b = 3.5$). Check: $3(4) - 2(3.5) = 12 - 7 = 5$ and $4 + 7 = 11$. The value of b 's neighbour 3 is a trap; forgetting to divide by 4 gives 16; subtracting instead of adding the equations gives 7.

60. (B) The fraction left is $1 - 1/3 - 1/4 = (12 - 4 - 3)/12 = 5/12$ of the total. So $(5/12) \times \text{total} = 250 \rightarrow \text{total} = 250 \times 12/5 = 600$. Check: rent 200, food 150, left 250, sum 600. Mis-adding the spent fractions to $1/2$ leaves $1/2 \rightarrow \text{total } 500$; arithmetic slips give 750 or 450.